

Data Scientist at the biggest bank in Russia and Computer Science student at Tsinghua University. Leveraging *4+ years of Machine Learning experience, including 3+ years as a Data Scientist and 1+ year as a Researcher*, to solve complex problems using AI tools

EXPERIENCE

Sberbank - Data Scientist (Multi-Agent Systems & NLP)

Oct 2025 — Present

- Engineered a multi-agent system (SGR, structured output, supervisor-executor) that assists project leaders in decomposing projects into actionable tasks while proactively identifying risks and recommending mitigation guidelines
- Developed an asynchronous multi-agent system that delivers *unbiased product analysis days faster than human experts*
- Designed an LLM-based pipeline (LangChain, function calling) that provides teams with best practice recommendations to enhance operational efficiency. *Expected outcome: average 10% reduction in lead time*

Tsinghua University - AI-Researcher (Multimodal AI & Multi-Agent Systems)

Aug 2024 — Oct 2025

- Enhanced the robustness and efficiency of multimodal information processing systems (*~2.7% increase in accuracy*) by integrating advanced DL-based pipelines (weighted fusion networks, modality reconstruction, Swin, Wav2vec, BERT)
- Built and deployed an MVP of a multi-agent system (AutoGen, FastAPI, Docker) for the repository documentation generation

VTB Bank - Data Scientist (NLP & Time Series Analysis)

Sep 2023 — Aug 2024

- Developed a pipeline for clustering (BERT, TF-IDF, SVD, K-Means) and hierarchical classification (LightGBM) of text requests received by technical support. The approach *reduces the average processing time of a request by ~25%*
- Built a time-series anomaly detection method (isolation forest, Prophet, volatility analysis). Deployed the pipeline (Airflow, Docker, PostgreSQL), enabling proactive identification of information system issues *hours ahead of human monitoring*
- Created a time series classifier (50+ feature generation, LightGBM) and integrated it with forecasting and anomaly detection tasks

Sberbank - Data Scientist (Classic ML & Recommender Systems)

Mar 2022 — Sep 2023

- Fully responsible for 15+ ad hoc projects* for multi-format retail businesses, forecasting existing store performance and ranking profitable expansion locations (PySpark, CatBoost, linear regression, spatial machine learning and feature engineering)
- Implemented a demand forecasting system (CatBoost, Prophet, Hyperopt) for sausage factory products. The pipeline estimates the volume of production and supply of goods, *enhancing the supply volume efficiency by ~20%*
- Analyzed the pairwise interactions of grocery products, and proposed a new approach to personalize recommendations

SKILLS

- Frameworks & Languages:** Python, PyTorch, Hugging Face Transformers, Claude, LangChain, LangGraph, AutoGen, scikit-learn, Gradient Boosting, Git, Docker, FastAPI, Airflow, Kubernetes, asyncio, multithreading, SQL, PostgreSQL, Bash, PySpark, $\LaTeX$
- AI-Domain Expertise:** NLP, LLM, Multi-Agent Systems, Multimodal AI, Classic ML, Time Series Analysis (Forecasting, Anomaly Detection), AI in Finance, ML model deployment, MLOps, Ranking, Recommendation Systems, Econometrics, Data Analysis
- Communication:** English (fluent), Russian (native), Chinese (beginner)

RESEARCH & PROJECTS

Research

- Speaker** at the **ODS AI** conference. Topic: Anomaly Scoring for Preventive Detection of Failures in Information Systems
- Author of the **paper** about a **Multi-Agent System** that improves accessibility by generating context-aware modality conversions
- Fine-Tuned** an LLM and **created** an AI-powered system that assigns transparency-focused reliability scores to web information
- Author of the **survey paper** about embedding AI into network devices to improve efficiency, latency, and topology optimization

Pet Projects

- Studied and applied modern NLP architectures and tooling: vectorizers (TF-IDF, fastText) for text clustering, transformer-based models (BERT, GPT) fine-tuning (vanilla, LoRA) for medical data, prompt engineering (few-shot, CoT, SGR)
- Took part in a Kaggle Clicks prediction competition (Vowpal Wabbit, PySpark). Ranked top 10%
- Applied machine learning methods to optimize the option price modeling and avoid the Black-Scholes model limitations

EDUCATION

Tsinghua University - Master's in Computer Science

Sep 2024 - Jun 2026

- Relevant courses: NLP, Information Retrieval and Recommender Systems, Time Series Intelligence, Machine Learning, Reading and Writing Technical Papers, Combinatorics and Algorithms, Computer Networks, Chinese

HSE University - Bachelor of Economics (Machine Learning & AI in Finance)

Sep 2020 - Jun 2024

- Teaching assistant for the Machine Learning in Macroeconomics course for graduate students
- Researcher at the Laboratory of Stochastic Analysis and its Applications. Topics: option price modeling using AI, neural networks
- Relevant courses: Deep Learning, NLP, Large-Scale Machine Learning, Machine Learning 1, Machine Learning 2, Data Analysis in Python, Probability Theory and Statistics, Stochastic Models, Econometrics, Financial Economics, Financial Markets